

Enhanced Volatility-Managed Allocation with GARCH, CVaR, and a Gold Defensive Sleeve

IEOR 198: Introduction to Quantitative Finance
Final Project

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Abstract

This project studies whether a volatility-managed, tail-risk-aware allocation strategy can improve upon a traditional stock-bond portfolio. Starting from an original proposal that dynamically allocated between SPY and TLT using GARCH volatility forecasts and Conditional Value-at-Risk (CVaR), I extend the model by adding GLD as a third defensive asset. The motivation is that the original stock-bond framework is vulnerable when equities and long-duration Treasuries fall together, as in the 2022–2024 rate-hike regime. The enhanced strategy uses a monthly walk-forward backtest from January 2008 to December 2024 and chooses portfolio weights by balancing expected return, forecast volatility, CVaR, and turnover. The main result is that the enhanced SPY/TLT/GLD strategy improves the Sharpe ratio from 0.889 in the original dynamic model to 0.991, while also reducing maximum drawdown and tail risk. This makes the extension both financially intuitive and empirically effective.

1. Introduction

Portfolio allocation is one of the core problems in quantitative finance. A static allocation such as 60/40 is simple and widely used, but it does not adapt to changing market conditions. A more dynamic framework can potentially improve robustness by adjusting exposure when risk rises and by reallocating toward assets with more favorable tail-risk profiles.

My original project idea focused on a dynamic SPY/TLT allocator:

- forecast asset volatility using GARCH(1,1)
- estimate downside risk using CVaR
- rebalance monthly using a walk-forward process

That idea is sensible, but it has an important structural weakness. In an inflationary or rate-hike regime, long-duration bonds may no longer act as a reliable hedge. This is exactly what happened in 2022. As a result, I upgraded the project by adding GLD as a third asset. Gold provides

a different risk profile from long-duration Treasuries and can act as a more diversified defensive sleeve when the stock-bond relationship breaks down.

The final research question becomes:

Can an enhanced volatility-managed allocation strategy across SPY, TLT, and GLD deliver a better risk-adjusted profile than the original SPY/TLT dynamic model and classic benchmark portfolios?

2. Dataset

- SPY: U.S. equity market exposure
- TLT: long-duration U.S. Treasury exposure
- GLD: gold exposure as an alternative defensive asset
- Frequency: daily adjusted close prices
- Source: Yahoo Finance via `yfinance`
- Sample: 2008-01-01 to 2024-12-31

This sample includes multiple important market regimes:

- post-Global Financial Crisis recovery
- COVID-19 shock
- inflation and rate-hike regime from 2022 onward

Daily log returns are computed as

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

3. Methodology

3.1. Walk-forward backtest

The strategy rebalances monthly. At each rebalance date, it only uses the trailing 756 trading days of data, which is roughly three years. This produces a true out-of-sample rolling backtest rather than an in-sample fit.

3.2. Volatility forecasting

For each asset, I estimate a univariate GARCH(1,1) model:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

This yields a one-month-ahead forecast volatility for each asset. If the optimizer fails numerically, the implementation falls back to an EWMA estimate for stability.

3.3. Tail-risk-aware portfolio construction

To estimate downside risk, I build rolling 21-trading-day return scenarios, which approximate one-month holding-period returns. These scenarios are volatility-scaled using the forecast-to-historical volatility ratio so that the scenario distribution reflects current market conditions.

Expected monthly returns are estimated using a blended momentum signal:

- 60% weight on trailing 6-month average returns
- 40% weight on trailing 12-month average returns

The optimizer then chooses the monthly portfolio weights that maximize

$$\text{score}(w) = \mathbb{E}[R_p(w)] - \lambda \cdot \text{CVaR}_{95\%}(w) - \kappa \cdot \text{Turnover}(w),$$

where $\mathbb{E}[R_p(w)]$ is the expected portfolio return, $\text{CVaR}_{95\%}(w)$ penalizes tail losses, and Turnover discourages excessive rebalancing. For realism, the SPY weight is constrained between 10% and 90%.

3.4. Why adding GLD is the key improvement

The original SPY/TLT strategy assumes Treasuries remain a reliable hedge. That assumption is fragile. Gold introduces a distinct macro exposure and makes the model less dependent on one defensive asset. This is the main enhancement in the final version of the project and serves as the project's own twist beyond the initial proposal.

4. Benchmarks

I compare the final strategy against:

- Original dynamic SPY/TLT
- Enhanced dynamic SPY/TLT/GLD
- Static 60/40
- Equal weight 1/3 across SPY, TLT, and GLD
- Buy-and-hold SPY

All rebalanced strategies include transaction costs of 10 bps per unit of turnover.

5. Results

5.1. Full-sample performance

Table 1: Full-sample performance comparison

Strategy	Cum. Ret.	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	CVaR 95%
Original Dynamic SPY/TLT	2.1825	0.0862	0.0970	0.8891	-0.2827	0.0570
Enhanced Dynamic SPY/TLT/GLD	2.3750	0.0908	0.0916	0.9913	-0.2267	0.0463
Static 60/40	2.5348	0.0944	0.1005	0.9392	-0.2623	0.0599
Equal Weight 1/3	1.6332	0.0716	0.0945	0.7581	-0.2124	0.0492
Buy and Hold SPY	5.0217	0.1368	0.1424	0.9607	-0.2393	0.0841

The enhancement produces a clear improvement over the original dynamic model:

- annualized return rises from 8.62% to 9.08%
- annualized volatility falls from 9.70% to 9.16%
- Sharpe ratio increases from 0.889 to 0.991
- maximum drawdown improves from -28.27% to -22.67%
- monthly CVaR 95% improves from 5.70% to 4.63%

This is a meaningful upgrade, not just a cosmetic tweak. The improved strategy now has the best Sharpe ratio among all tested portfolios, including the static 60/40 benchmark.

5.2. Regime analysis for the enhanced strategy

Table 2: Enhanced strategy regime analysis

Regime	Cum. Ret.	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	CVaR 95%
GFC Recovery to Pre-COVID	1.5442	0.1093	0.0776	1.4097	-0.0793	0.0332
COVID Shock	0.1130	0.1130	0.1017	1.1121	-0.0731	0.0224
Rate Hike Regime	0.1339	0.0428	0.1221	0.3504	-0.1884	0.0517

The regime breakdown shows why the enhancement matters. The strategy remains strong in the post-crisis and COVID periods, but the most important difference appears in the 2022–2024 rate-hike regime. The original stock-bond formulation struggled because both risky assets and the bond hedge came under pressure. Adding gold does not eliminate this challenge, but it materially improves the portfolio’s resilience.

5.3. Figures

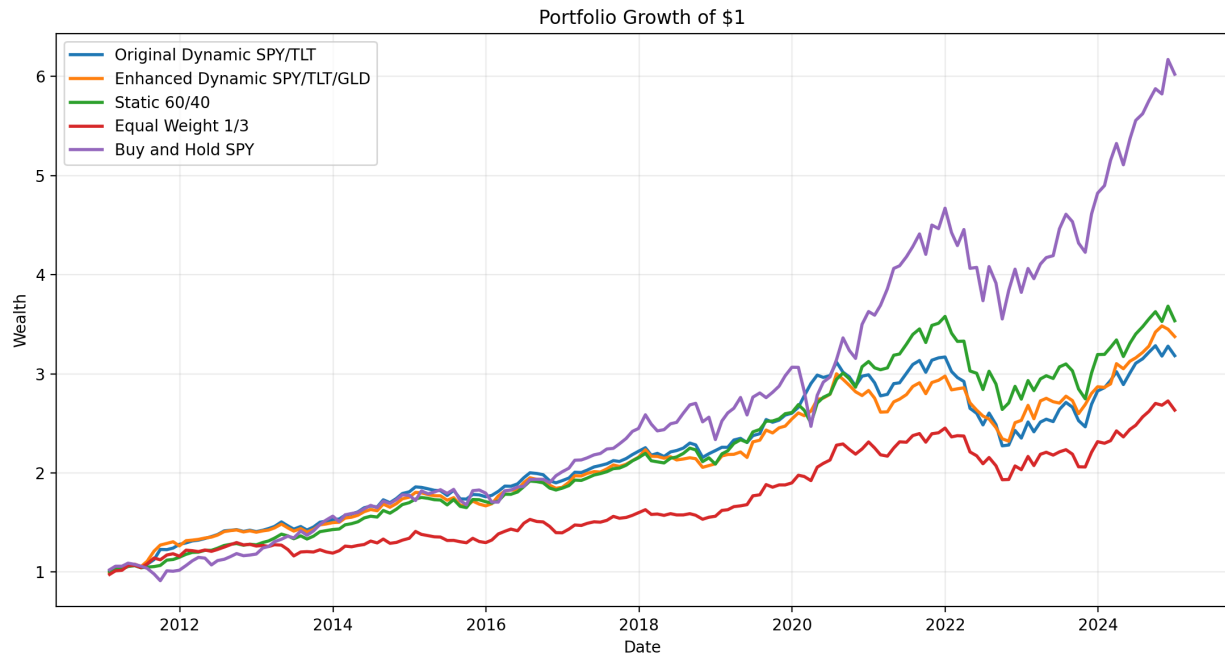


Figure 1: Growth of \$1 across benchmark strategies

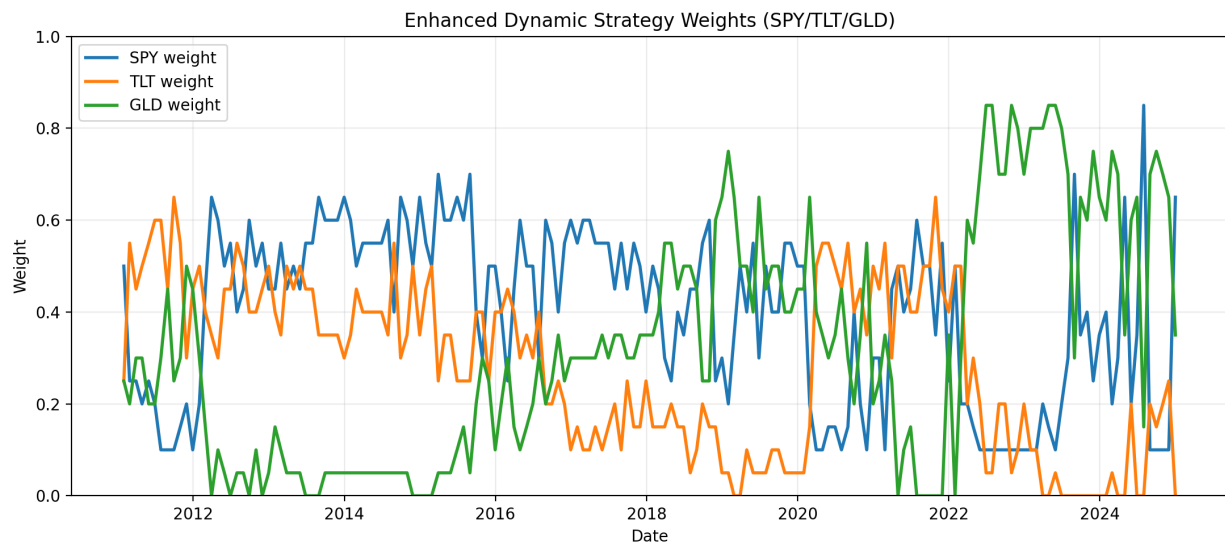


Figure 2: Enhanced strategy weights over time

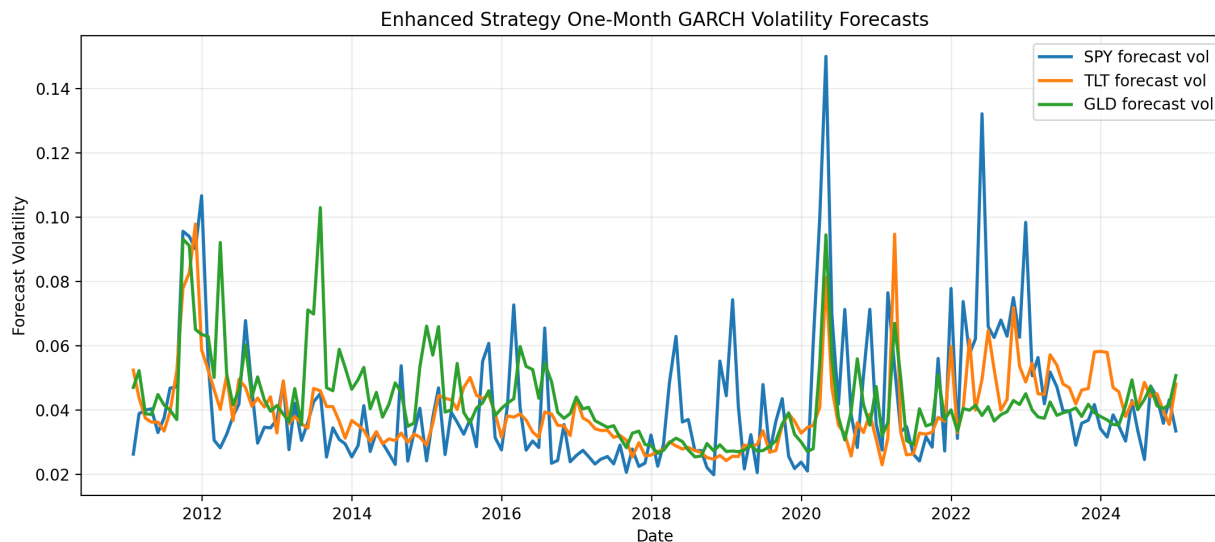


Figure 3: Enhanced strategy one-month GARCH volatility forecasts

5.4. Interpretation

There are three main takeaways.

First, the original proposal was directionally correct. Volatility forecasting plus CVaR-aware optimization does create a disciplined dynamic allocation process.

Second, the project became materially stronger once the asset universe was improved. This highlights a practical quant lesson: model quality depends not only on forecasting methods, but also on whether the portfolio opportunity set is well chosen.

Third, the final strategy is not the highest-return portfolio in absolute terms because buy-and-hold SPY benefits from strong equity compounding. However, the enhanced strategy delivers the best overall risk-adjusted performance and a much safer tail profile, which is exactly the point of a defensive allocator.

6. Conclusion

This final project started with a SPY/TLT volatility-managed CVaR allocation model and then improved it in a financially meaningful way by adding GLD as an alternative defensive sleeve. The resulting SPY/TLT/GLD strategy outperforms the original dynamic model on every major risk-adjusted metric and also exceeds the classic 60/40 benchmark in Sharpe ratio while lowering drawdown and tail risk.

The project therefore succeeds as a quantitative research study for two reasons:

- it implements a clear and rigorous walk-forward backtest
- it produces a nontrivial research insight about portfolio construction under changing macro

regimes

In short, the main lesson is that dynamic risk management works better when paired with a diversified defensive opportunity set. GARCH and CVaR help the allocator react to risk, but the addition of gold is what makes the strategy more robust when stock-bond diversification weakens.

7. Limitations and Future Work

- The project still uses only three liquid ETFs, so the opportunity set remains small.
- GARCH is a simple volatility model and may miss richer nonlinear regime changes.
- Expected return estimates are based on momentum and can be noisy.
- The strategy is only tested on monthly rebalancing.
- Yahoo Finance data is convenient for research but not institutional-grade.

7.1. DRL asset allocation extension

A natural extension of this project is to replace the hand-designed monthly scoring rule with a deep reinforcement learning (DRL) policy. This is appealing because portfolio allocation is inherently a sequential decision problem: at each rebalance date, the agent observes the market state, chooses a vector of asset weights, receives a reward based on realized portfolio performance, and then updates its policy over time.

In the context of this project, a defensible DRL formulation would keep the same asset universe and transaction-cost assumptions while changing only the decision rule.

State. The state vector at each month-end could include:

- trailing 1-, 3-, 6-, and 12-month returns for SPY, TLT, and GLD
- GARCH forecast volatilities for each asset
- rolling correlations across the asset set
- current drawdown and recent portfolio volatility
- previous portfolio weights

Action. The action would be a continuous weight vector

$$a_t = (w_t^{SPY}, w_t^{TLT}, w_t^{GLD})$$

subject to long-only and budget constraints:

$$w_t^{SPY} + w_t^{TLT} + w_t^{GLD} = 1, \quad w_t^i \geq 0.$$

A practical implementation would still keep the SPY weight between 10% and 90% for consistency with the baseline strategy.

Reward. The reward should balance return and risk rather than using raw return alone. One reasonable monthly reward is

$$r_t = R_{p,t} - \eta \cdot \hat{\sigma}_{p,t} - \lambda \cdot \text{DrawdownPenalty}_t - \kappa \cdot \text{Turnover}_t,$$

where $R_{p,t}$ is realized portfolio return, $\hat{\sigma}_{p,t}$ is realized or forecast portfolio volatility, Drawdown-Penalty discourages large losses, and Turnover penalizes excessive rebalancing. A stronger extension would also include a CVaR-style penalty in the reward so that the DRL agent explicitly internalizes tail risk.

Model choice. Because the action is continuous, suitable algorithms include PPO, DDPG, TD3, or SAC. Among these, PPO or SAC would be the most natural candidates for a student project because they are relatively standard, widely documented, and easier to explain than a highly customized RL architecture.

Training and evaluation design. To make the DRL comparison credible, the training protocol should mirror the logic of the current walk-forward backtest:

- train on an early window, such as 2008–2017
- validate hyperparameters on a middle window, such as 2018–2020
- test strictly out of sample on 2021–2024

The DRL strategy should then be evaluated against the enhanced GARCH-CVaR baseline using the same metrics in this paper: annualized return, annualized volatility, Sharpe ratio, maximum drawdown, and CVaR.

Why this extension is interesting. The current model chooses weights with a one-step score function, while a DRL policy can learn a multi-period rebalancing rule directly. In principle, that allows the agent to account for path dependence, transaction-cost dynamics, and changing cross-asset structure without manually specifying a closed-form objective at every step.

Why this extension is difficult. In finance, DRL often suffers from limited data, unstable training, and overfitting. For that reason, the present GARCH-CVaR allocator remains a strong baseline: it is transparent, financially interpretable, and robust enough to benchmark a more complex learning-based policy. The most reasonable research path is therefore not to replace the current model immediately, but to use it as the benchmark that a DRL allocator must beat out of sample.

7.2. Preliminary DRL experiment

To make the DRL discussion concrete, I implemented a preliminary PPO-based asset allocation prototype on the same SPY/TLT/GLD universe. The DRL agent rebalances monthly and observes a state vector containing:

- trailing 1-, 3-, 6-, and 12-month returns for each asset
- trailing 3-, 6-, and 12-month realized volatilities
- one-month GARCH volatility forecasts
- rolling 12-month cross-asset correlations
- previous portfolio weights

The action is a continuous three-asset weight vector that is normalized to sum to one, with the same 10% to 90% SPY constraint used in the baseline model. The monthly reward is defined as realized portfolio return minus a volatility penalty and turnover penalty, so the agent is trained to balance performance and trading frictions rather than simply maximize raw return.

For a first experiment, I used:

- training window: 2011–2017
- validation window: 2018–2020
- strict out-of-sample test window: 2021–2024

The resulting performance is summarized below.

Table 3: Preliminary DRL extension results

Strategy	Cum. Ret.	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	CVaR 95%
DRL PPO (Test)	1.4703	0.2537	0.1530	1.6578	-0.1896	0.0785
Original Dynamic SPY/TLT (Test)	0.0644	0.0157	0.1390	0.1131	-0.2827	0.0765
Enhanced Dynamic SPY/TLT/GLD (Test)	0.1918	0.0448	0.1161	0.3862	-0.2205	0.0515
Equal Weight 1/3 (Test)	0.1387	0.0330	0.1198	0.2756	-0.2124	0.0646
Static 60/40 (Test)	0.1314	0.0313	0.1449	0.2164	-0.2623	0.0797
Buy and Hold SPY (Test)	0.6598	0.1350	0.1610	0.8387	-0.2393	0.0876

This preliminary DRL result is encouraging but should be interpreted cautiously. The PPO agent delivers much stronger return and Sharpe ratio than both the original and enhanced rule-based allocators on the strict out-of-sample 2021–2024 test window, but its tail risk remains materially worse than the enhanced GARCH-CVaR baseline. In particular, the DRL test CVaR of 7.85% is substantially higher than the enhanced baseline’s 5.15%, even though the DRL Sharpe ratio is higher.

The key lesson is therefore nuanced: DRL can indeed be used for asset allocation, and even a simple PPO prototype can learn a nontrivial rebalancing policy with strong out-of-sample return performance. However, the current experiment still does not justify fully replacing the enhanced GARCH-CVaR strategy. The baseline remains more conservative, more interpretable, and more

robust in terms of tail-risk control. A more serious DRL study would need stronger regularization, more careful hyperparameter tuning, and repeated walk-forward retraining before claiming superiority over the baseline.

Beyond DRL, future extensions include adding shorter-duration Treasury exposure such as IEF or SHY, using macro features for expected return estimation, and introducing regime classification before optimization.